

A Minor Project Final Report on
Real Estate Marketplace and Valuation
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ABSTRACT

This project addresses the information asymmetry and inefficient processes prevalent in the Nepali real estate market. By leveraging the MERN (MongoDB, Express.js, React, Node.js) stack, we have developed a comprehensive online real estate marketplace platform. The initial phase successfully established core functionalities, enabling users to efficiently search, list, and view properties with detailed information. Concurrently, a robust data collection effort was done, gathering diverse data sources to train a valuation model. Rigorous data cleaning and pre-processing was done to ensure the dataset's quality for accurate valuations.

The final output will be a user-friendly platform equipped with an advanced valuation model, empowering users with data-driven insights. Additionally, the platform will offer features like advanced search filters, communication tools and seamless communication, aiming to transform the Nepali real estate market by enhancing transparency, efficiency, and user experience.

Keywords: online platform, comprehensive property database, property management system, data collection, data cleaning, pre-processing, Elastic Net regression, machine learning, valuation model, multicollinearity, feature selection

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1. INTRODUCTION

In today's digital age, online platforms have revolutionized various industries, including real estate. The real estate market in Nepal is evolving and with the increasing digitization of various industries, there's a growing demand for online platforms that cater to the needs of property brokers.

To address this need, we propose the development and launch of an online real estate marketplace focusing primarily on the seller side. Our project aims to revolutionize the real estate market by creating a user-friendly, transparent, and efficient online platform where buyers, sellers, and real estate agents can seamlessly interact. This will foster a sense of trust and reliability among users, making our platform the go-to destination for real estate transactions in Nepal. We envision this project not just as a marketplace but as a catalyst for the modernization of the Nepali real estate industry, fostering transparency, efficiency, and growth. This document outlines the project's scope, features, technical specifications, and overall development plan.

1.1 PROBLEM STATEMENT

The Nepali real estate market is experiencing a period of rapid growth and transformation, yet property brokers face significant challenges in effectively serving clients and capitalising on market opportunities. A lack of access to a centralized, transparent database of property information impedes accurate property valuation, leading to missed opportunities for sellers and potential risks for buyers due to price discrepancies. This further increases the existing information asymmetry in the market, fuelling inefficiencies and hindering fair transactions.

Also, the absence of a robust property management system tailored to the Nepali market poses additional challenges. Brokers struggle to efficiently manage multiple property listings, communicate effectively with clients, and streamline the transaction process. This lack of streamlined tools results in missed leads, delayed deals, and a suboptimal experience for both brokers and their clients

1.2 PROJECT OBJECTIVE

- To create a comprehensive and reliable source for property valuation based on property information including details on listings, transactions and market pricing, that empowers brokers with accurate data for valuation and informed decision-making.
- To provide brokers with a streamlined platform to efficiently manage listings, communicate with clients, automate routine tasks, and track transactions, ultimately enhancing productivity and client satisfaction.

1.3 SCOPE AND LIMITATIONS

Here is an overview of the scope of our property marketplace and valuation system:

1.3.1 SCOPE:

- **Property Listing:** The system will maintain a detailed database of property listings, including property type, location, size, amenities, images, pricing history and ownership details.
- **Search and Filtering:** Users will be able to search for properties based on various criteria, such as location, price range, property type, and specific features.
- **Property Valuation Tool:** The system will incorporate a robust valuation model, leveraging elastic net regression and a wide range of data points, to generate accurate and reliable property valuations for both brokers and clients.
- **Listing Management:** Brokers can easily create, edit, and manage property listings, including uploading images, videos, and other relevant documents.
- **Communication Tools:** Integrated messaging and email functionalities will enable seamless communication between brokers, clients, and other stakeholders.
- **Mobile Accessibility:** The system will be optimized for mobile devices to provide a seamless user experience on smartphones and tablets.

1.3.2 LIMITATIONS:

The development of a web-based real estate platform in Nepal, while offering significant opportunities, faces certain limitations and challenges due to the unique characteristics of the Nepali real estate market:

- **Lack of Centralized Database:** The absence of a comprehensive, centralized real estate database in Nepal poses a major challenge. Data is often fragmented, inconsistent, and scattered across various sources, making it difficult to ensure accuracy and completeness.
- **Inconsistent Data Sources:** Property information is often sourced from diverse entities such as local agents, brokers, and individual sellers, leading to discrepancies in data formats, accuracy, and timeliness.
- **Complex Regulations:** Nepal's real estate regulations are multifaceted and can vary significantly across municipalities, posing challenges for compliance and standardization within the platform.
- **Land Ownership Issues:** Land ownership records can be outdated, incomplete, or disputed, creating complexities in property verification and transaction processes.
- **Informal Transactions:** A substantial portion of real estate transactions in Nepal occur informally, without proper documentation or adherence to legal procedures. This informal market poses challenges in capturing accurate and reliable data for the platform.
- **Economic Fluctuations:** Nepal's economy is susceptible to fluctuations, impacting property values and demand. This necessitates continuous monitoring and updating of market data on the platform to maintain relevance and accuracy.
- **Internet Penetration:** While internet usage is increasing, there are still regions in Nepal with limited or unreliable internet access, hindering the reach and usability of the web-based platform for a wider audience.

- **Technology Adoption:** There can be resistance to adopting new technologies, particularly among older generations or those accustomed to traditional real estate practices. This necessitates user education and intuitive platform design to encourage adoption.
- **Building Trust:** Establishing trust in a market prone to fraud and misinformation is paramount. This requires stringent verification of property listings, agent credentials, and robust security measures to protect user data.
- **Localization:** Tailoring the platform to Nepali language, cultural preferences, and local measurement standards is crucial for user engagement and adoption. This includes adapting to local currencies, terminology, and navigation patterns.

2. LITERATURE REVIEW

The proposed electronic real estate marketplace discussed in the research paper “The Impact of Internet Real Estate Intermediary Platform on the Real Estate Market” (by authors Zhang, Wenzhan & Chen, Shaofei & Guo, Dongmei & Li, Bing) [1]. Zhang et al. (2019) delve into the impact of internet real estate intermediary platforms for the real estate market [1]. They explore how these platforms, which have surged in popularity, affect property transactions, pricing dynamics, and overall market efficiency. The authors highlight the transformative power of these platforms in increasing market transparency. By granting buyers and sellers easy access to vast amounts of information—property listings, historical prices, neighbourhood data—they level the playing field, empowering consumers to make informed decisions [1]. However, the authors acknowledge potential downsides, such as furthering inequalities due to limited internet access or digital literacy among some populations [1]. These also include concerns about data accuracy, the need for robust security measures to protect sensitive information and the potential for online platforms to disrupt traditional real estate agent roles. Overall it is found that online platforms have positively transformed the way properties are bought and sold.

The research paper titled “Comparative evaluation of the performance of different regression models in land valuation” by authors Yalprı, S., & Yalprı, E. present a comparative analysis of various regression models for land valuation, emphasizing the growing importance of machine learning (ML) algorithms in this domain [2]. The study focuses on predicting land values in central neighbourhoods of Konya, Turkey, using Lasso, Elastic Net, ML.Net, and Ordinary Least Squares (OLS) regression models. The results reveal that ML.Net, a Microsoft-developed framework for building custom ML models, exhibits superior performance compared to other models, including the commonly used OLS [2]. However, the study is limited to a specific geographic location and dataset, raising questions about the generalizability of its findings to other contexts [2]. Further research across diverse regions and datasets is needed to validate the superior performance of ML.Net and explore its potential in broader land valuation applications.

The research paper titled “Locally Regularized Linear Regression in the Valuation of Real Estate” by author Kubus, Mariusz [3], explores a novel approach to real estate valuation using locally regularized linear regression. In this study, the author challenges the conventional practice of building a single global regression model for property valuation. Instead, Kubus proposes the use of local regression models, which are tailored to specific subsets of data, offering greater flexibility and potentially improved accuracy in capturing local market variations [3]. Kubus conducted empirical research using three real estate market datasets to test the effectiveness of locally regularized linear regression [3]. The results showed promising outcomes, with the local models outperforming global models in terms of prediction accuracy [3].

The research paper “Land Valuation and Management Issues in Nepal” by authors Shrestha, R.L., and Rajbhandari, K. presented at the FIG (International Federation of Surveyors) Workshop in Nepal, addresses the critical issues surrounding land valuation and management in the country [4]. It highlights the absence of a unified, standardized system for land valuation, leading to inconsistencies and inaccuracies in property assessments. The current practice often relies on subjective judgment and outdated methods, resulting in disputes and undervaluation of land, which can hinder economic development and equitable land distribution [4]. It highlights the urgent need for reforms and adoption of modern technologies to establish a robust and equitable land valuation system in the country.

Laalpurja Nepal is a prominent online real estate marketplace operating in Nepal, catering to buyers, sellers, and renters across the country [5]. The platform offers a comprehensive listing of properties, including residential, commercial, and land options, along with detailed descriptions, images, and contact information for agents or owners. While Laalpurja Nepal has established itself as a popular platform, its impact on the Nepali real estate market and its overall effectiveness in addressing the challenges of the industry warrant further research. A comprehensive analysis of user experiences, transaction success rates, and the platform's influence on property prices and market trends would provide valuable insights into its role and potential for improvement.

3. METHODOLOGY

3.1 Project Methodology

The project team has opted to use an iterative development strategy to build the system in stages. This approach offers flexibility, adaptability, and the ability to deliver value early in the development cycle. It allows for manageable increments, each delivering a functional piece of the system. This approach reduces project risks, incorporates early user feedback, and allows for changes in requirements as the project progresses.

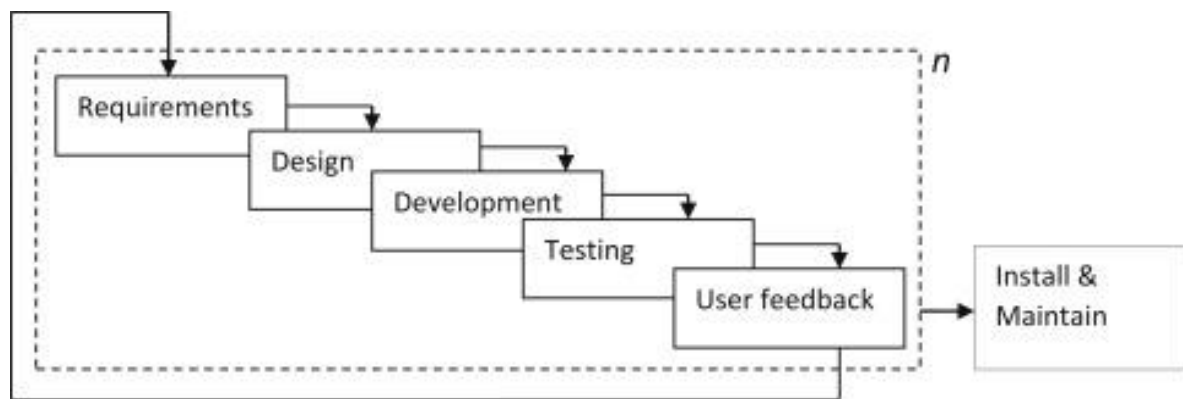


Figure 1: Incremental Model

Increment 1:

1. Planning and Requirements Gathering:
 - o Defined the overall project scope and objectives.
 - o Prioritized features and functionalities based on user needs and business value.
 - o Conducted thorough market research and competitor analysis.
 - o Gathered detailed requirements for the initial increment
2. Design and Development:
 - o Designed the system architecture, database schema for the entire project.
3. Testing and Deployment:
 - o Conducted rigorous testing (unit, integration, and system testing) to ensure quality and functionality.

- o Gathered user feedback and incorporate necessary adjustments before deploying to the production environment.
 - o Deployed the first increment of the system to a staging environment for further testing and feedback.
4. Subsequent Increments:
- o Prioritized increments based on user feedback, market trends, and business goals.
 - o Enhanced the property database, refined the valuation model, and expanded the property management system features in each increment.
 - o Continuously tested and deployed each increment to ensure seamless integration with existing functionalities.

Increment 2:

1. Planning and Requirements Gathering:
 - o Gathered requirement for core functionalities like basic property listings and search using filter.
2. Design and Development:
 - o Implemented the initial front-end interface for user interaction.
 - o Developed the core backend functionalities, such as property listing creation, search, and filtering.
3. Testing and Deployment:
 - o Conducted rigorous testing (unit, integration, and system testing) to ensure quality and functionalities

Increment 3:

1. Planning and Requirements Gathering
 - o Gathered required resources having the data needed for the property valuation model
 - o Designed a pipeline for the implementation of the valuation model
2. Design and Development
 - o Scraped multiple resources to get relevant data

- o Data cleaned and processed
 - o Integrated the valuation model (Elastic Net Regression) using relevant data sources
3. Testing and Deployment
 - o Conducted testing (unit, integration, and system testing) to ensure quality and functionality

Increment 4:

1. Planning and Requirements Gathering
 - o Used algorithm to test valuation model
2. Design and Development
 - o Broke the data into K number of folds
 - o Tested the different model output
3. Testing and Deployment
 - o Conducted testing (unit, integration, and system testing) to ensure quality and functionality

3.2 Diagrams

3.2.1 ER-Diagram of Property Marketplace and Valuation

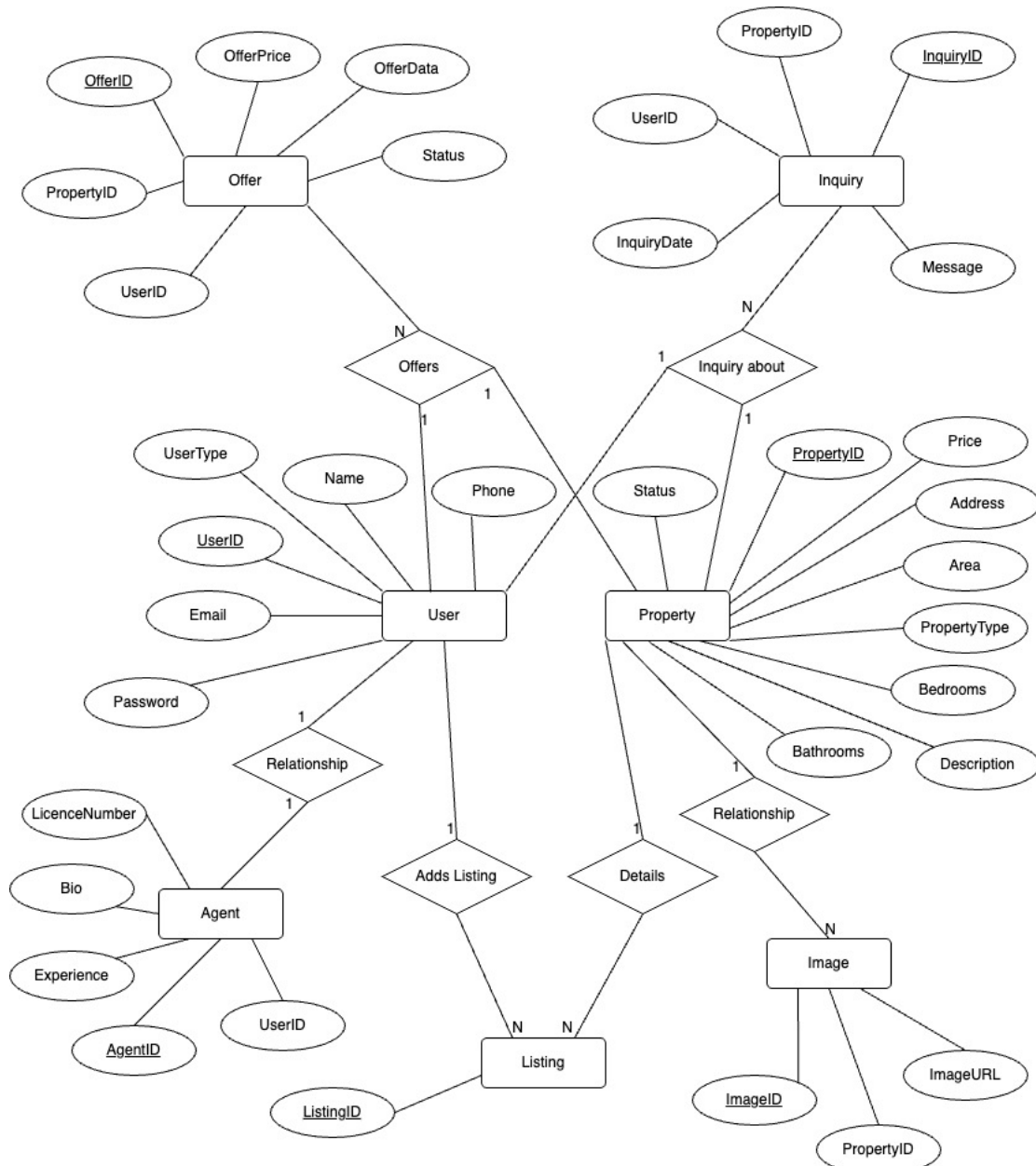


Figure 2: ER diagram of Property Marketplace and Valuation

Entities:

- **User:** This is the central entity, representing users of the platform. It has attributes such as UserID, Name, Email, Password, UserType (to differentiate between buyers, sellers, and agents), Phone. Additionally, agents have specific attributes like License Number, Bio, and Experience.
- **Property:** Represents real estate properties listed on the platform. Attributes include PropertyID, Price, Address, Area, PropertyType, Bedrooms, Bathrooms, and Description.
- **Listing:** Connects properties with users who add them to the platform. It also has a relationship with Images, allowing multiple images per listing.
- **Image:** Stores image details like ImageID and ImageURL for each property listing.
- **Offer:** Records offers made on properties. It has attributes like OfferID, OfferPrice, OfferDate, and Status. It also has a relationship with the User who made the offer and the Property the offer is for.
- **Inquiry:** Captures inquiries about properties. It has attributes like InquiryID, InquiryDate, and Message, along with relationships to the User who made the inquiry and the Property being inquired about.

Relationships:

- **User and Property:** A user can add many listings, and a listing is added by one user. This is a one-to-many relationship.
- **User and Agent:** A user can be an agent, but an agent is also a user. This is a one-to-one relationship using inheritance.
- **Listing and Property:** Each listing refers to one property, and a property can have many listings. This is another one-to-many relationship.
- **Listing and Image:** A listing can have many images. This is another one-to-many relationship.
- **User and Offer:** A user can make many offers, and an offer is made by one user. This is a one-to-many relationship.

- Offer and Property: An offer is made on one property, and a property can have many offers. This is another one-to-many relationship.
- User and Inquiry: A user can make many inquiries, and an inquiry is made by one user. This is a one-to-many relationship.
- Inquiry and Property: An inquiry is about one property, and a property can have many inquiries. This is another one-to-many relationship.

3.2.2 Use Case Diagram of Property Marketplace and Valuation

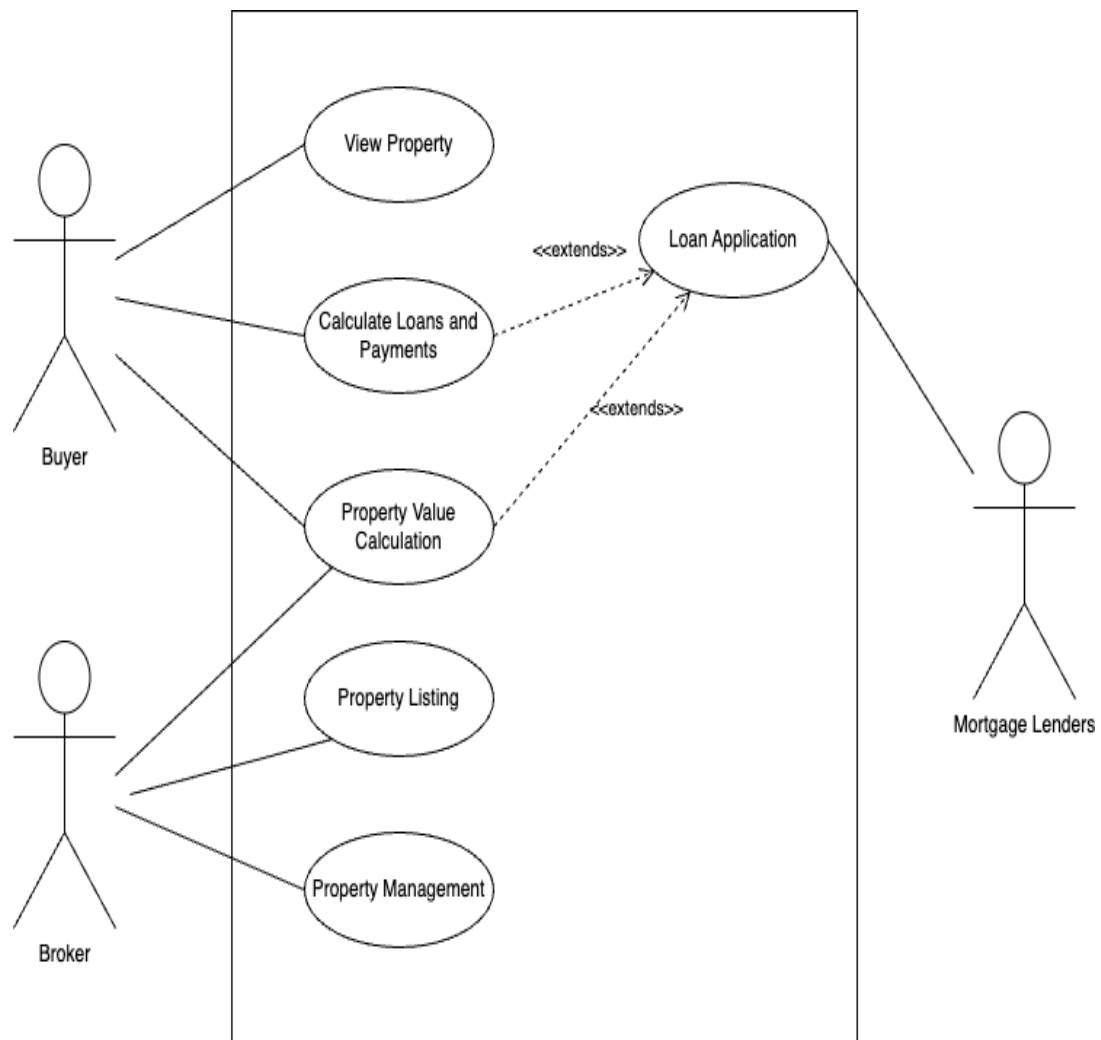


Figure 3: Use Case diagram of Property Marketplace and Valuation

Actors:

- Buyer: The individual or entity interested in purchasing a property.
- Broker: The real estate professional facilitating property transactions.
- Mortgage Lender: While not directly shown, their involvement is implied through the "Loan application" use case.

Use Cases:

- View Property: Buyers can browse and view property listings on the platform.
- Calculate Loan and Payments: Buyers can use a built-in calculator to estimate loan amounts and monthly payments based on property prices and their financial situation.
- Loan Application: Buyers can apply for a mortgage loan through the platform, potentially interacting with mortgage lenders (though this detail isn't explicit in the diagram).
- Property Value Calculation: This could represent the platform's automated valuation model (AVM), providing estimates of property values to both buyers and brokers.
- Property Listing: Brokers can list properties for sale on the platform.
- Property Management: Brokers can manage their property listings (update details, mark as sold, etc.)

3.2.3 Activity Diagram of Property Marketplace and Valuation

3.2.3.1 Activity Diagram Marketplace

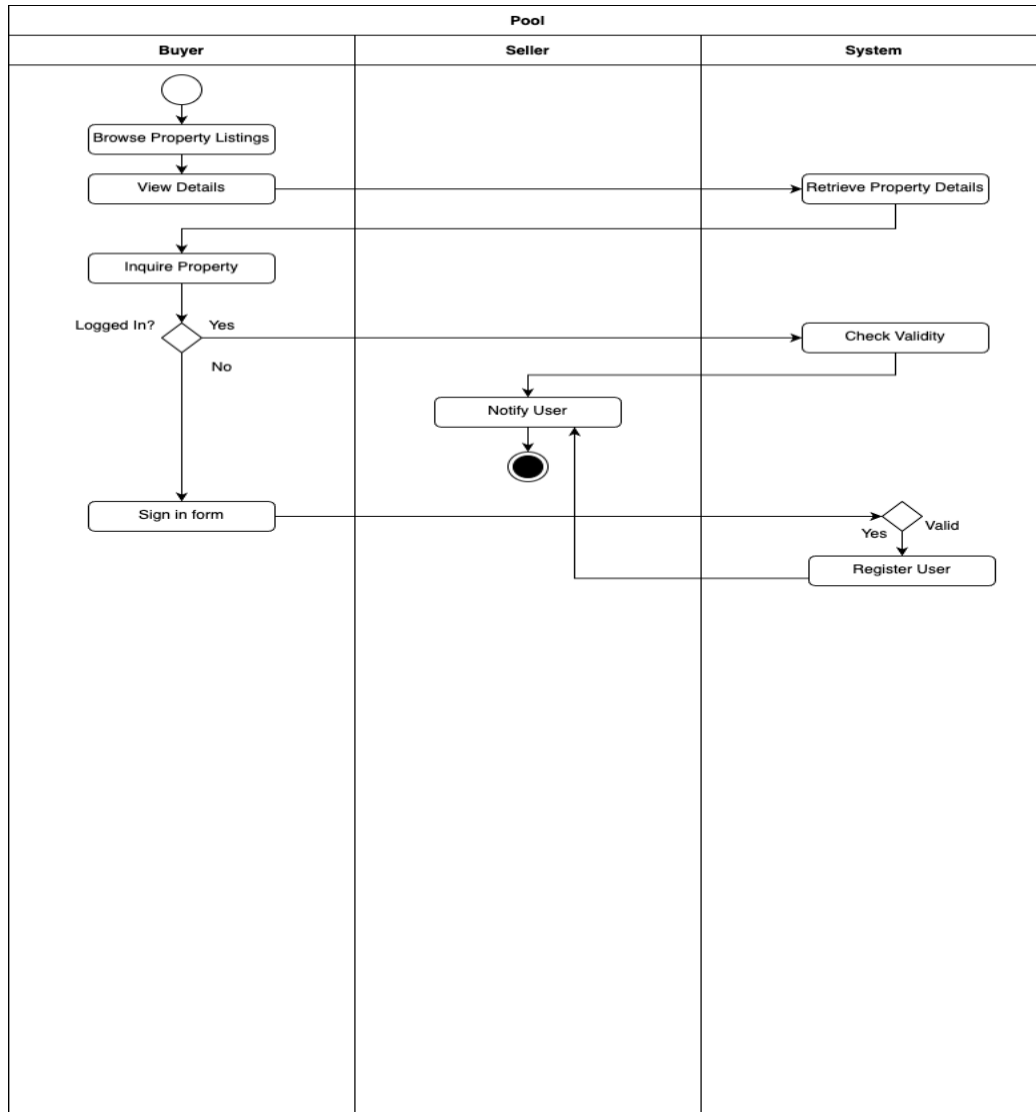


Figure 4: Activity Diagram (Marketplace)

3.2.3.2 Activity Diagram of Valuation Model

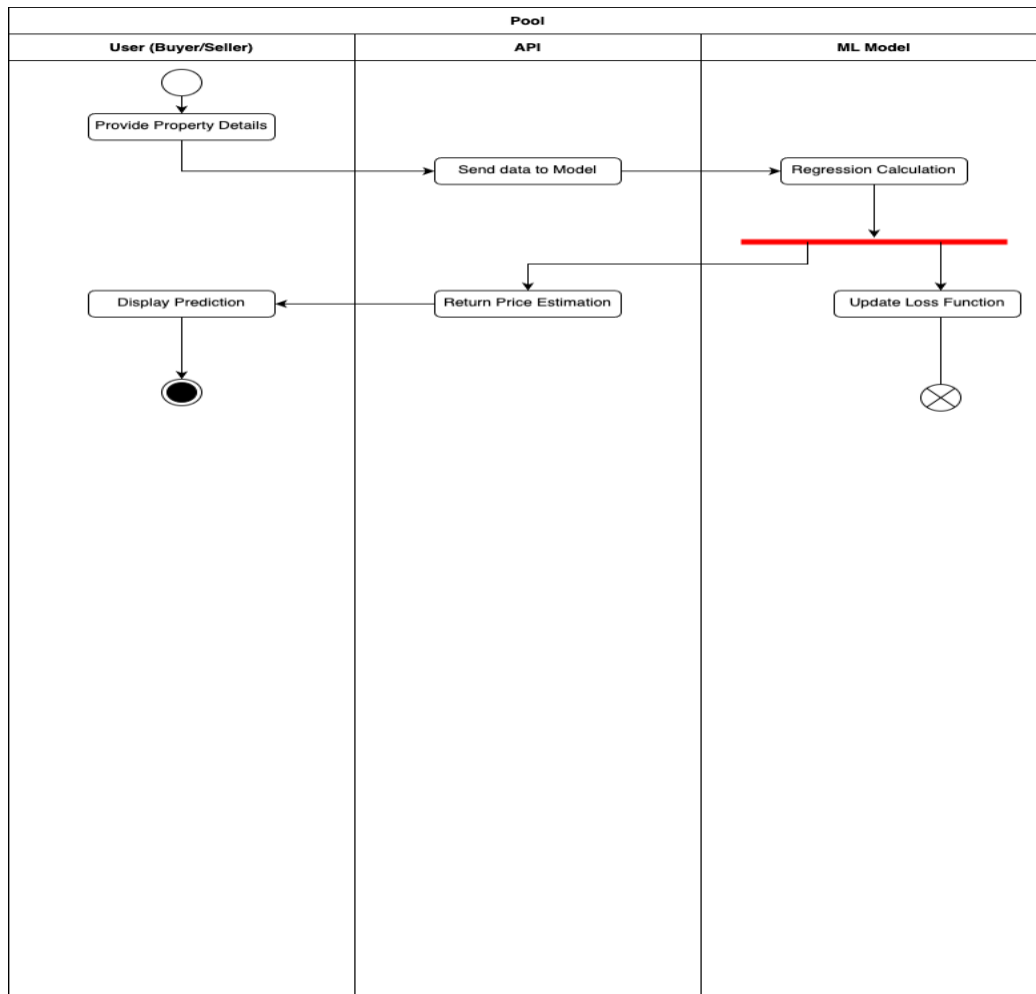


Figure 5: Activity Diagram (Valuation Model)

3.2.4 Data Flow Diagram of Property Marketplace and Valuation

3.2.4.1 Level 0 Data Flow Diagram

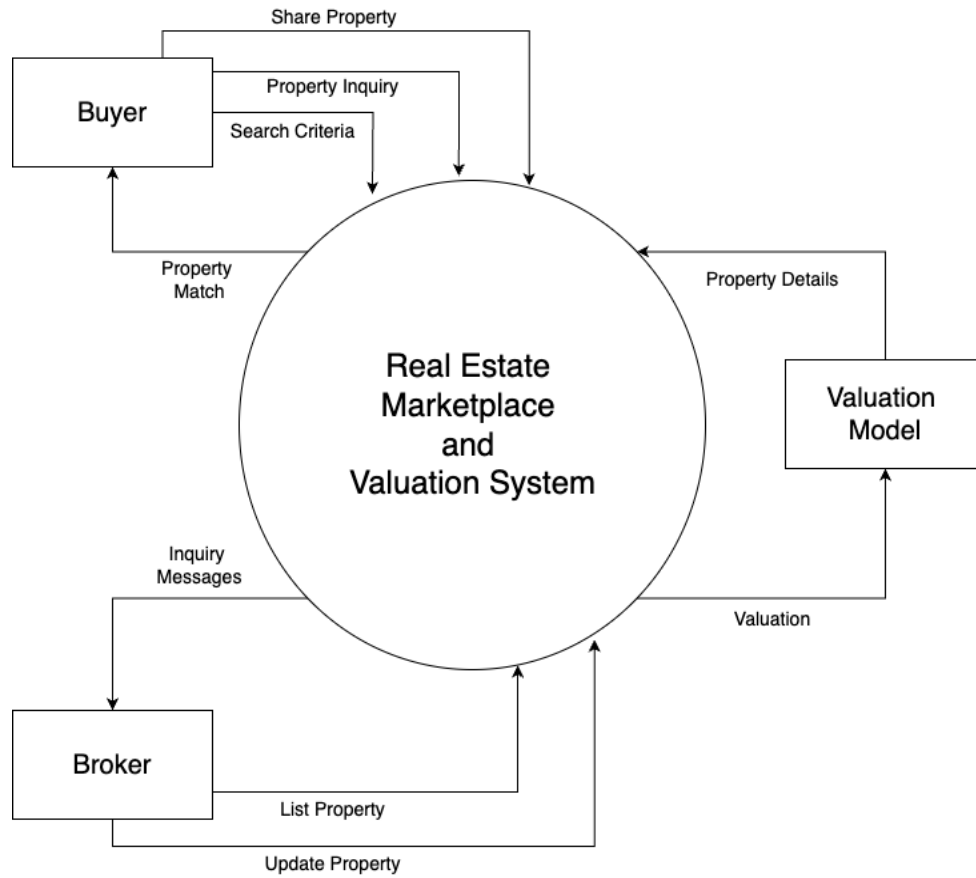


Figure 6: Level 0 Data Flow Diagram of Marketplace and Valuation

3.2.4.2 Level 1 Data Flow Diagram

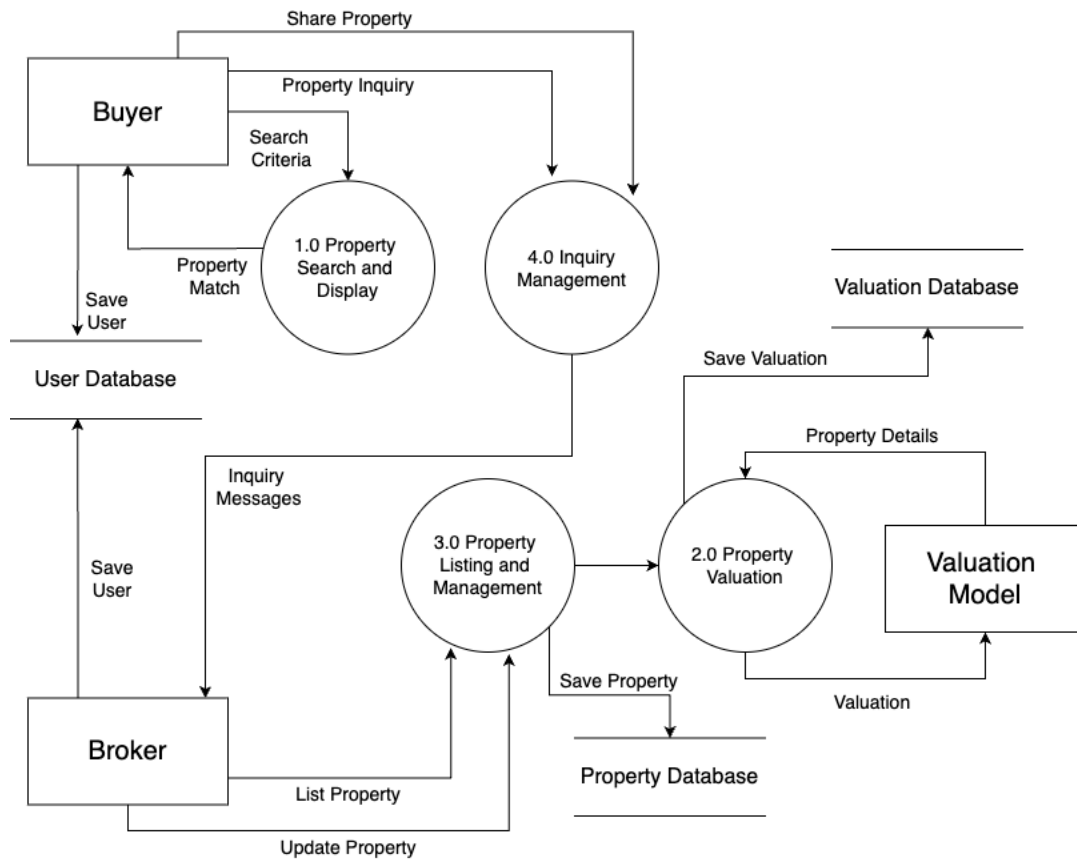


Figure 7: Level 1 Data Flow Diagram of Marketplace and Valuation

3.2.5 Component Diagram of Property Marketplace and Valuation

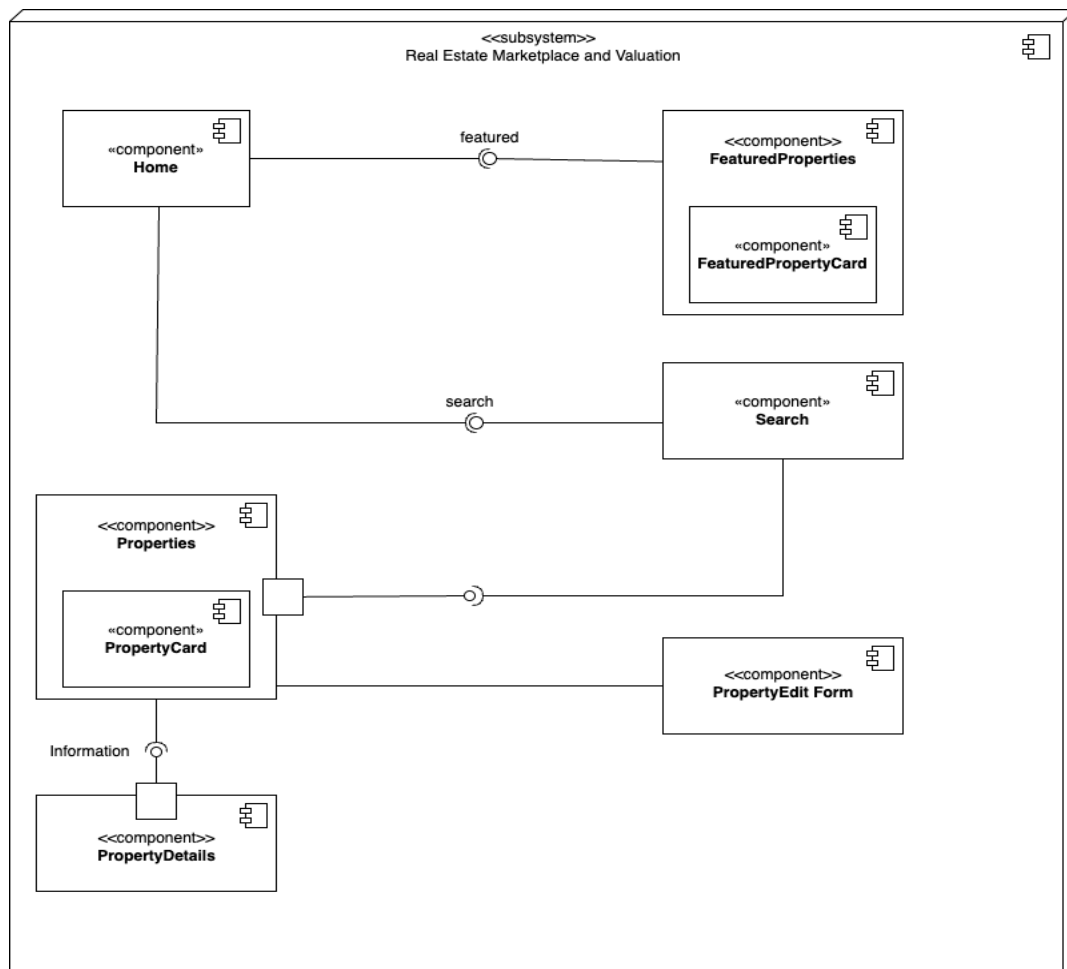


Figure 8: Component Diagram of Property Marketplace and Valuation

3.2.6 Deployment Diagram of Property Marketplace and Valuation

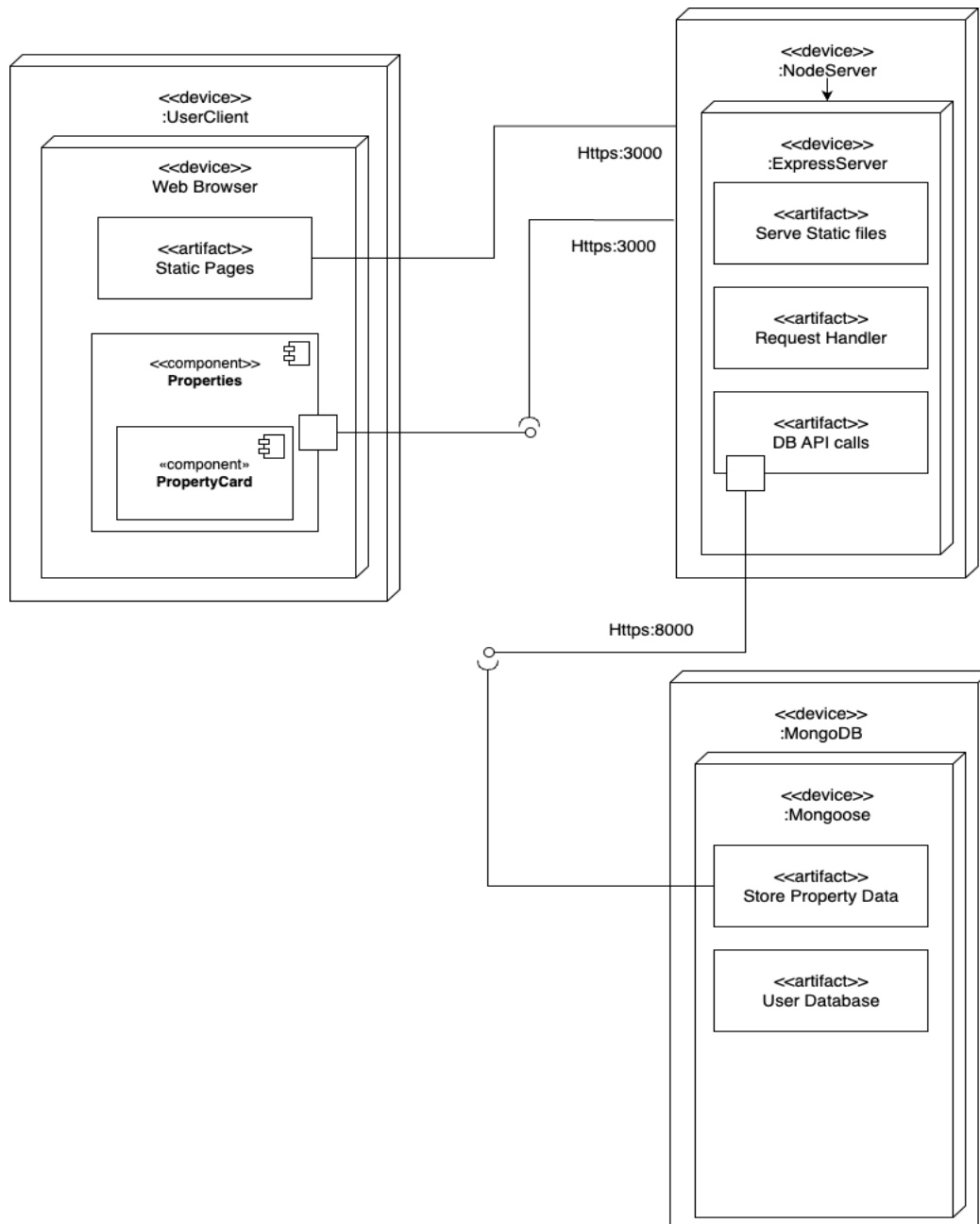


Figure 9: Deployment Diagram of Property Marketplace and Valuation

3.2.7 Sequence Diagram of Property Marketplace and Valuation

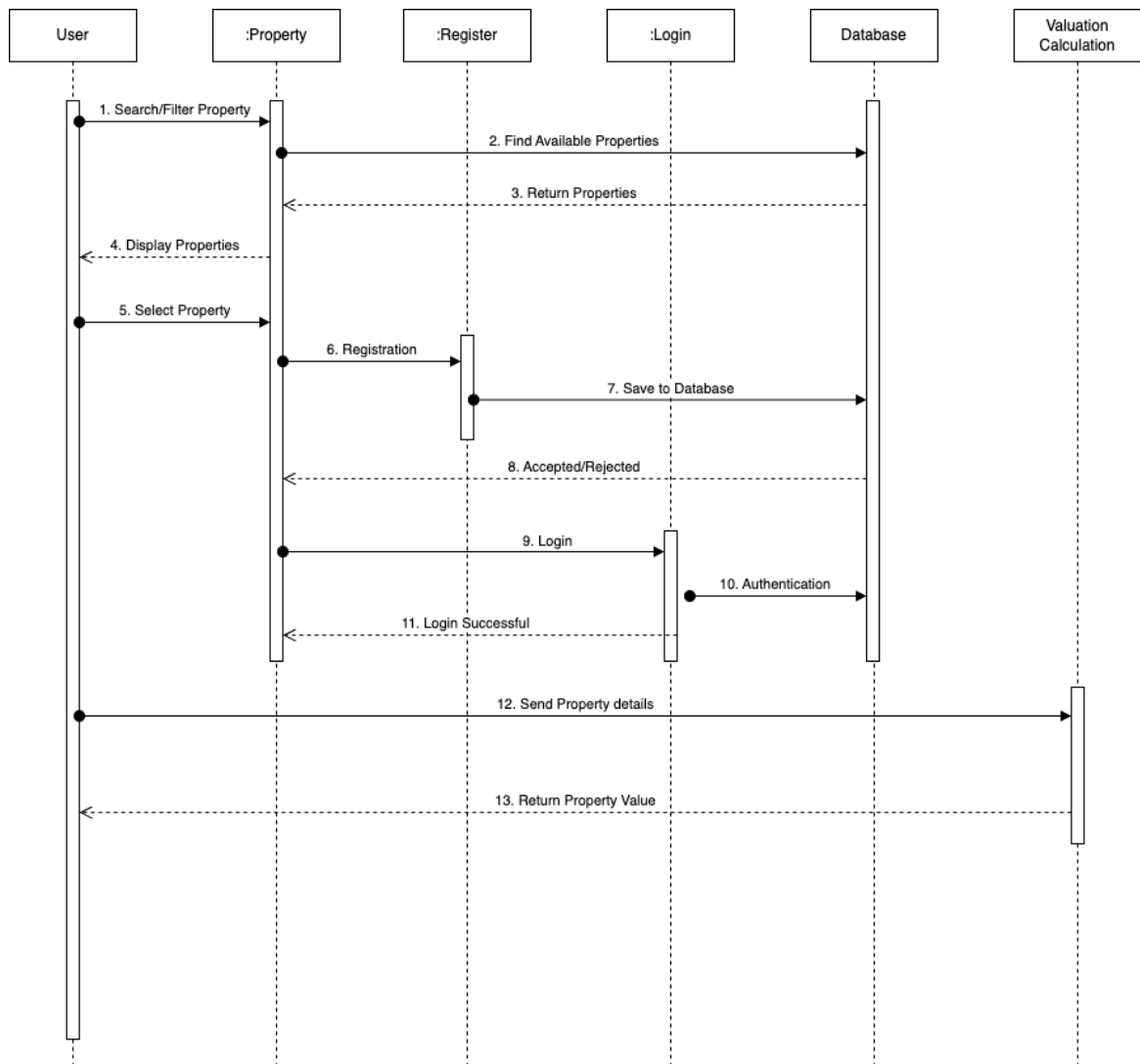


Figure 10: Sequence Diagram of Property Marketplace and Valuation

3.3 Algorithms Used

3.3.1 Z Scaling

Z-score standardization, also known as standard scaling, transforms each feature to have a mean of 0 and a standard deviation of 1. This is achieved through a simple formula:

$$Z = (X - \mu) / \sigma \quad \text{—————} \quad (1)$$

Where:

- Z is the standardized Z-score
- X is the original feature value
- μ (mu) is the mean of the feature across the entire dataset
- σ (sigma) is the standard deviation of the feature

In the development of our real estate project's Valuation Model, a crucial step in the data pre-processing phase involved the utilization of Z-scores as a scaling mechanism. This standardization technique played a pivotal role in ensuring the model's accuracy and fairness by mitigating the impact of varying feature scales.

3.3.2 K means clustering

K-means clustering is an unsupervised machine learning algorithm that iteratively partitions a dataset into 'K' distinct clusters, where each data point belongs to the cluster with the nearest mean (centroid).

The algorithm was applied to property location data (latitude and longitude) along with relevant features like area size in Aana, and price in crore. To gain a visual understanding of the identified clusters, we utilized mapping tools and data visualization libraries.

Thus, K-means clustering was leveraged to identify patterns and groupings within property data, specifically focusing on geographic location and property attributes.

3.3.3 Linear Regression

Linear regression is a supervised learning algorithm that models the relationship between a dependent variable (the target we want to predict) and one or more independent variables (features or predictors) by fitting a linear equation to the observed data.

In our real estate project, linear regression served as one of the fundamental machine learning algorithms employed in the development of our Valuation Model. This widely used statistical technique enabled us to establish a linear relationship between property characteristics (features) and their corresponding sale prices (target variable), ultimately facilitating accurate property valuations.

3.3.4 Elastic Net Regression

This is a regularized regression technique that combines the strengths of Lasso and Ridge regression, effectively handling multicollinearity and feature selection, which are common issues in real estate data.

$$\min_{\beta_0, \beta} \frac{1}{2N} \sum_{i=1}^N (y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \left[(1 - \alpha) \frac{1}{2} \sum_{j=1}^p \beta_j^2 + \alpha \sum_{j=1}^p |\beta_j| \right] \quad (2)$$

where:

- o y_i is the observed property value for the i -th observation
- o β_0 is the intercept term
- o β_j are the coefficients for the predictor variables x_{ij}
- o λ is the regularization parameter controlling the overall strength of regularization
- o α is mixing parameter, balancing the contributions of L1 and L2 penalties ($0 \leq \alpha \leq 1$)

3.4 Tools and Technology Used

3.4.1 FRONTEND

3.4.1.1 HTML

We have used HTML for the structure and organization needed to present content effectively on the web. We specify attributes within HTML tags to customize the appearance and behaviour of elements, such as headings, paragraphs, links etc.

3.4.1.2 CSS (Tailwind)

We have used Tailwind CSS to style and design our web pages which differ from traditional CSS frameworks by focusing on a utility-first approach. It provides a comprehensive set of utility classes that can be applied directly in the HTML markup, allowing us to quickly style elements without writing custom CSS rules. Responsiveness was achieved by applying responsive breakpoints and using utility classes that hide, show, or modify elements based on the viewport width.

3.4.1.3 JavaScript

We have used JavaScript to add interactivity, dynamic behaviour, and functionality to our web pages. JavaScript is a versatile programming language that runs on the client side (in the web browser) and enables us to create interactive elements, manipulate the document object model (DOM), handle user events, and perform various tasks on webpages. JavaScript provided access to various web APIs (Application Programming Interfaces) that offer additional functionality.

3.4.1.4 NEXT JS

We have used Next.js which is a React framework developed by Vercel that simplifies building high-performance web applications. We opted to use Next as it offers features like server-side rendering (SSR), static site generation (SSG), automatic code splitting, optimized prefetching, and a streamlined developer experience. The framework simplifies development with built-in routing, API routes, and automatic optimization, while offering flexibility to choose rendering methods for different pages.

3.4.1.5 Material UI

We have used Material-UI to enhance the visual design and user experience of our web application. A key advantage of Material-UI is its ready-to-use UI components such as buttons, forms, navigation bars, cards, dialog boxes, and more, emphasized clean and modern aesthetics, consistent visual cues, and smooth interactions. By utilizing these pre-built components, we quickly and effectively build a visually appealing and cohesive user interface.

3.4.2 BACKEND

3.4.2.1 Node JS

We have used Node JS as it allows us to build high-performance applications with its efficient and scalable event-driven architecture. It leverages a unified JavaScript stack for both frontend and backend development, streamlining the process and reducing complexity. Also, Node.js boasts a vast ecosystem of open-source libraries and modules through npm, providing us with a wealth of resources

3.4.2.2 Express JS

We use Express.js as it is a minimalist and flexible web application framework for Node.js that simplifies the process of building web applications and APIs. It provides a robust set of features for handling routing, middleware, and templating, while remaining lightweight and unopinionated.

3.4.2.3 Mongo DB

MongoDB is used as our database which uses a document-oriented data model, storing data in flexible, JSON-like documents. This allows for easy storage of hierarchical data without a predefined schema, making it adaptable to evolving application requirements. It empowers developers to handle diverse data types with its flexible document-oriented model, adapting effortlessly to changing application needs.

3.4.3 Beautiful Soup

We used Beautiful Soup to extract data from HTML and XML files. It provides us with a simple and intuitive way to navigate, search, and modify the parse tree of a parsed document. We primarily used it to fetch existing property listing data

3.4.4 FAST API

Fast API is a high-performance web framework for building APIs with Python. Fast API is used to communicate interface our frontend with the machine learning model which gives the valuation of properties based on provided property information

3.4.5 Pandas Data Library

We used Pandas for data analysis and manipulation. It is used to easily manipulate and analyse data with its intuitive data structures and extensive functionality. It provides efficient tools for reading and writing data in various formats, including CSV and Excel. Its flexible indexing and slicing capabilities simplify data selection and filtering.

3.5 Data Collection

The data for this project was acquired through a multi-faceted approach, leveraging both web scraping techniques and manual aggregation of publicly available information.

3.5.1 Web Scraping:

- Primary sources: Data was extracted from two prominent real estate platforms in Nepal:
 - LaalPurja.com: A real estate broker specializing in property listings and transactions.
 - GharSansar.com: A real estate aggregator consolidating listings from various sources.
- Secondary sources: Additional data was scraped from smaller real estate websites to supplement the primary sources and enhance the diversity of the dataset.

3.5.2 Manual Data Aggregation:

- Publicly available data: Information regarding property transactions, prices, and other relevant details was manually collected from public sources such as government records, news articles, and real estate publications. This provided a broader perspective on the real estate market in Nepal.

3.6 Data Cleaning and Pre-processing

The data resulted from the collection efforts have amassed some amount of data from diverse sources that were heterogeneous in format and quality. Further, the data cleaning and pre-processing phase has revealed inconsistencies, missing values, and outliers in the raw data, underscoring the importance of this crucial step. Therefore, a systematic cleaning and pre-processing pipeline was implemented to ensure data consistency, accuracy, and suitability for analysis and model training.

3.6.1 Data Consolidation:

- All scraped data from various sources was consolidated into a single CSV (Comma-Separated Values) file, facilitating easier manipulation and analysis.

3.6.2 Feature Selection:

- Relevant columns were carefully selected based on their potential impact on property valuation and market analysis. These included:
 - Location value (city name)
 - Address (street plus city name)
 - Price
 - Price label (crore or lakhs)
 - Total area
 - Bathroom count
 - Bedroom count

3.6.3 Handling Missing Values:

- Empty rows with all columns missing were removed, as they provided no useful information.

- Columns with a significant number of missing values (bathroom and bedroom count) were imputed using appropriate techniques. Imputation methods could include using the mean, median, or mode of the respective columns, or employing more sophisticated algorithms like K-Nearest Neighbours (KNN) imputation.

3.6.4 Feature Extraction (Geocoding):

- The 'geopy' Python framework was utilized to extract latitude and longitude coordinates from the location data (city name and address). This enabled spatial analysis and visualization of property locations.

3.6.5 Feature Engineering

- Price Conversion:

Price data was standardized by converting all values to crores (e.g., 25,000,000 to 2.5) based on the price label (crore or lakhs), ensuring uniformity and comparability across listings.

- Area Conversion:

The total area, originally presented in various units (ropani, square feet), was converted to a standardized unit of measurement, aana, to ensure consistency and facilitate analysis.

The data cleaning and pre-processing pipeline ensured that the final dataset was accurate, reliable, and ready for subsequent analysis and model training. By addressing missing values, standardizing units, and enriching the data with geographic coordinates, we created a robust foundation for the development of our real estate valuation model

3.7 Data Analysis and Modelling

The data analysis process for this project involved a series of steps to derive meaningful insights from the collected real estate data, encompassing exploratory data analysis (EDA), and statistical analysis.

The project's methodology also encompassed a comprehensive data modelling approach to develop a robust Valuation Model that enhances the Real Estate Marketplace.

3.7.1 Exploratory Data Analysis (EDA)

The initial phase involved a thorough exploration of the cleaned and pre-processed dataset. This included:

- **Data Profiling:** Understanding the dataset's structure, size, and variable types through summary statistics (mean, median, mode, range) and frequency distributions.
- **Data Visualization:** Utilizing various plots (scatter plots, histograms, bar charts) to visualize the relationships between variables, identify trends, and detect potential outliers.
- **Outlier Detection and Treatment:** Employing statistical techniques (e.g., Z-score, IQR) to identify and address outliers to prevent them from skewing the analysis and modelling results.

3.7.2 Statistical Analysis:

This stage focused on quantifying relationships and patterns within the data:

- **Correlation Analysis:** Calculating correlation coefficients to assess the strength and direction of linear relationships between different features and the target variable (property price).
- **Hypothesis Testing:** Formulating and testing hypotheses to determine the statistical significance of observed relationships and make inferences about the model results.

3.7.3 Neighbourhoods Identification (K-Means Clustering):

Visualizing the resulting clusters gave us insights into the spatial distribution of property values and neighbourhoods characteristics.

- Clustering: Applying K-means clustering to property location data (latitude, longitude) and other relevant features (e.g., property type, size, price) to group similar properties into distinct clusters representing neighbourhoods.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

4.1.1 Real Estate Marketplace

- User Management:
 - User registration, login, and profile management.
 - User roles (buyers, sellers, agents) with differentiated functionalities.
- Property Listings:
 - Ability to create, edit, and manage property listings.
 - Comprehensive property details including photos, videos, virtual tours (optional).
 - Search functionality with filters based on location, price, property type, and other relevant criteria.
 - Listing reviews and ratings (optional).
- Communication Tools:
 - Secure messaging system between buyers, sellers, and agents.
 - Inquiry management for buyers to contact sellers/agents.

4.1.2 Valuation Model

- Data Acquisition:
 - Ability to collect and integrate data from various sources (e.g., public records, MLS listings, user-submitted data).
 - Data cleaning and pre-processing functionalities to ensure data quality.
- Model Training:
 - Training algorithms to generate property valuations based on historical data and market trends.
 - Support for different valuation models (e.g., statistical models, machine learning models).
- Valuation Generation:
 - Ability to generate property valuations based on user-provided property details.
 - Option to display valuation confidence intervals or explainability tools to understand the reasoning behind the valuation.
- Model Maintenance:
 - Mechanisms to update the model with new data and monitor its performance over time.

4.2 Non-functional Requirements

4.2.1 Performance and Scalability

- The platform should handle a high volume of users and concurrent connections.
- The system should be responsive and provide fast loading times for listings and search results.
- The valuation model should be able to generate valuations within a reasonable timeframe.
- The platform should be scalable to accommodate future growth in users and listings.

4.2.2 Security

- Secure user authentication and authorization mechanisms.
- Encryption of sensitive user data (e.g., financial information).
- Regular security audits and penetration testing to identify and address vulnerabilities.
- Secure communication protocols to protect data transmission.

4.2.3 Availability and Reliability

- The platform should be highly available with minimal downtime.
- Redundancy measures to ensure system functionality even during hardware or software failures.
- Disaster recovery plan to address potential data loss or outages.

4.2.4 Usability

- User-friendly interface for both buyers, sellers, and agents.
- Intuitive search and navigation functionalities.
- Accessibility features for users with disabilities.
- Clear and concise instructions for using the platform.

4.2.5 Maintainability

- The codebase should be well-documented and modular for easy maintenance and future enhancements.
- Use of industry-standard coding practices and frameworks.
- Version control system for tracking changes and facilitating collaboration.

5. TESTING

5.1 Test Plan

The purpose of the test plan is to communicate the intent of the testing activities with end user representatives, stakeholders, system integrator, implementer and tester. Test cases are conducted with the last version. Functional and non-functional requirements are tested with test cases. All the requirements are listed and tested according to the priority. For instance, “Essential” requirement has higher priority than “Desired” requirements and “optional” requirement. Higher Priority requirements are tested before lower priority requirements.

5.2 Test strategy

The test strategy presents the recommended approach for testing a part of a system of system as a whole.

5.2.1 Unit Testing

Unit testing ensures that the part of the program works well. It is performed by implementer at the time of implementation. The implementer checks values for the return types. They will also validate the different tools boxes used like Text View, Edit Text etc.

5.2.2 Integration Testing

The components in the implementation model are integrated and tested to ensure proper operation. One bigger working package will be integrated to other smaller working packages in incremental manner.

5.2.3 System Testing

System testing will be conducted by tester after the completion of construction phase. After well-formed subset of the use case behaviour is implemented, all the functional and non-functional requirements are tested.

5.3. Test Cases

Step	Action	Expected System Response	Pass/Fail
1	Click on view properties	The system will display the list of properties and platform for text field, filter and Search button.	Pass
2	On entering the keyword or location or using filter and Click on Search	The system will display the filtered list of properties according to search query.	Pass
3	Click on Property valuation	The system will display form to add property details field.	Pass
4	On Entering the Property Details and click on Get Property Value button	The system will display the predicted valuation.	Pass
5	Add Property before Logging in	Adding property cannot be accessed without logging in	Fail
6	Add Property after Logging in	The system will show form asking to provide the details	Pass
7	Edit property details	System provides the platform for editing in details	Pass
8	Click on Send Message before login	Messaging cannot be accessed without logging in	Fail
9	Click on Send Message after login	The system will pop up the message “Message sent”.	Pass
10	Click Login and Registration	System direct to google login and registration system	Pass
11	Click sign out button	The system displays a message of successful log out	Pass

12	On click on notification bell	System will display the list of different notification	Pass
13	Bookmark Property after login in	The system will pop up the message “bookmarked added”.	Pass
14	Bookmark Property before login in	Bookmark cannot be accessed without logging in	Fail
15	Sharing properties	System navigates to the targeted media.	Pass
16	Click on profile icon	The system will display the user profile view	Pass

Table 1: Test Case

5.4 Test Results

The test coverage results as follows:

Total Test Case	Successful Test Case	Failed test case
16	13	3

Table 2: Test Result

Out of 16 test cases, 7 test cases passed and remaining 3 test cases failed so some them are listed for consult. Failure is found to be related Bookmarking, Adding property, Messaging without login.

6. DELIVERED OUTPUT

The delivered outputs of our Real Estate Marketplace and Valuation projects are:

1. A responsive User Interface and seamless interaction. [Figure 11]
2. Browse all listings with advanced Search and Filtering functionalities. [Figure 12]
3. Add custom property listings for registered brokers. [Figure 13]
4. View single listing and comprehensive property data. [Figure 14]
5. Track properties of interest through bookmarking properties. [Figure 15]
6. Contact seller/agent directly through in app messaging channels. [Figure 16]
7. Enter property details for property valuation. [Figure 17]
8. Valuation of your property from our model. [Figure 18]
9. REST API for calculating property value.

By delivering these outputs, we aim to create a valuable resource for the Nepali real estate market, empowering users with the information and tools they need to make informed decisions and facilitating efficient and transparent transactions.

7. RESULT AND DISCUSSION

7.1 Real Estate Marketplace

The development phase has yielded a functional real estate marketplace platform with core features that streamline property search, listing, and viewing. Preliminary user testing has generated positive feedback, indicating the platform's intuitive interface and ease of use. Test users have expressed appreciation for the comprehensive property listings, detailed information, and high-quality images, which have significantly enhanced their property search experience.

The user's openness to utilizing the marketplace for property listing and buying signifies its potential to address the existing information gap and inefficiencies in the traditional offline market.

Discussion:

The initial results of the project demonstrate the potential of the online marketplace platform to transform the Nepali real estate market. By providing a centralized platform with comprehensive property information, efficient search functionalities, and accurate valuations, we can empower users, increase market transparency, and streamline transactions.

However, the project is still in its early stages, and further research and development are needed to fully realize its potential. The implementation of the valuation model and the continuous expansion of platform features will be critical milestones in achieving the project's long-term goals. Additionally, ongoing efforts to collect and refine data will ensure the model's accuracy and adaptability to the dynamic real estate market.

Overall, this project represents a significant step towards digital transformation and modernization of the Nepali real estate sector. The preliminary findings offer promising insights into the potential impact of online platforms and machine learning techniques in addressing the existing challenges and unlocking new opportunities in the market.

7.2 Real Estate Valuation Model

The development of an accurate and reliable Valuation Model for the Nepalese real estate market involved a rigorous process of experimentation, analysis, and refinement. This section details the results obtained at each stage of model development and the subsequent discussions that informed our decisions.

1. Linear Regression

- **Initial Model:** A basic linear regression model was trained using the pre-processed and scaled dataset. The model's performance was evaluated using the following metrics:

Metric	Training Set	Test Set
Mean Squared Error (MSE)	0.7944587352158975	0.923371854795044
Mean Absolute Error (MAE)	0.691504948040444	0.691504948040444
R-squared	0.1977660808008761	0.08355186123475944

Table 3: Linear Regression Metrics

Discussion: While the model achieved a relatively lower MSE (0.794) on the training set, it exhibited a higher MSE (0.923) on the test set, indicating a degree of overfitting. Similarly, the R-squared value was higher on the training set (0.198) compared to the test set (0.084), suggesting that the model's ability to explain the variance in property prices is weaker on unseen data.

The MAE values for both training and test sets were identical (0.692). This metric provides a more interpretable measure of the average absolute difference between predicted and actual prices. In this case, the model's predictions were off by an average of 0.692 crores on both the training and test sets.

Overall, the results highlight the limitations of the linear regression model in capturing the complexities of the Nepalese real estate market. The model's tendency to overfit and its relatively low explanatory power on unseen data underscore the need for exploring alternative models that can better generalize to new property listings.

2. Gradient Descent Optimization

- **Cost Function Contour Plot:** A contour plot of the cost function (MSE) with respect to the model's coefficients was generated. This visualization aided in understanding the error landscape and the trajectory of gradient descent.
- **Gradient Descent Implementation:** The gradient descent algorithm was applied to iteratively optimize the model's parameters and minimize the MSE loss.

The Gradient Descent Optimization resulted in the following

Parameter	Value
Learning Rate	0.0001
Number of Iterations	1000
Final Weights (w)	[0.41994663, -0.03518224, -0.05029491]
Final Intercept (b)	1.9376841625265738e-16
Final Loss (MSE)	0.8181901446508015

Table 4: Gradient Decent Weights and MSE

Discussion: The linear regression model was trained using gradient descent optimization with a learning rate of 0.0001 and 1000 iterations. The resulting model achieved a final Mean Squared Error (MSE) of 0.8181901446508015. This indicates that, on average, the squared difference between the predicted property values and the actual values was 0.818. The model's weights (w) and intercept (b) represent the learned coefficients of the linear equation that best fits the relationship between property features and prices. However, as discussed previously, the linear regression model's performance was not optimal, prompting the exploration of alternative models like ridge, lasso, and polynomial regression.

3. Regularized Regression (Ridge and Lasso)

- **Ridge Regression:** Ridge regression, a regularization technique that adds a penalty term to the loss function, was applied to mitigate potential overfitting.
- **Lasso Regression:** Lasso regression, another regularization technique, was also employed. It differs from ridge regression in the type of penalty applied, potentially leading to feature selection.

The model's performance was evaluated and compared:

Metric	Training Set	Test Set
Ridge Regression		
Mean Squared Error (MSE)	0.7944594683504602	0.9230330042329634
Mean Absolute Error (MAE)	0.6789817077733339	0.7378273073341078
R-squared	0.1977653404912949	0.08388817099482604
Lasso Regression		
Mean Squared Error (MSE)	0.9903080837182895	1.0376186271380796
Mean Absolute Error (MAE)	0.7693964142438235	0.7963735134815233
R-squared	0.0	-0.02983825492483594

Table 5: Ridge and Lasso Regression Performance Metrics

Discussion: Both ridge and lasso regression, as regularization techniques, were applied to address potential overfitting in the linear regression model. However, their performance and impact on feature selection differed significantly.

- Ridge Regression: Ridge regression demonstrates a negligible improvement in R-squared on the test set compared to linear regression. While the MSE on the test set is slightly lower for ridge regression, the MAE is higher. This suggests that ridge regression, while potentially reducing some overfitting compared to linear regression, does not significantly enhance the model's predictive performance or generalizability in this context.
- Lasso Regression: Lasso regression performs worse than both linear and ridge regression on all metrics. The R-squared values are 0 on the training set and negative on the test set, indicating that the model does not explain any of the variance in property prices and may even be performing worse than simply predicting the mean. The higher MAE and MSE values further confirm the poor performance of lasso regression in this scenario.

4. Model Diagnostics

- **Rainbow Test:** This test was conducted to assess the linearity assumption of the models.

Statistic	Value
Rainbow Test Statistic	1.4589353226059576
P-value	0.00020876601470701925

Table 6: Rainbow Test

Discussion: The Rainbow Test, a statistical test for assessing linearity, was applied to evaluate the appropriateness of a linear model for our real estate data. The resulting test statistic of 1.4589 and the extremely low p-value (0.0002) provide strong evidence to reject the null hypothesis of linearity.

- **Breusch-Pagan Test:** This test was used to check for heteroscedasticity (unequal variances) in the residuals.

Test	P-value
Breusch-Pagan Test	3.647115712982391e-38

Table 7: Breusch-Pagan Test

Discussion: The Breusch-Pagan test, a statistical test for homoscedasticity (constant variance of errors), was applied to assess the validity of the assumption of equal variance in our regression models. The resulting p-value of 3.647e-38 is extremely low, providing strong evidence to reject the null hypothesis of homoscedasticity.

The rejection of the null hypothesis suggests the presence of heteroscedasticity in our data, meaning that the variance of the errors (residuals) is not constant across different levels of the predictor variables. This implies that the accuracy of our model's predictions may vary for different ranges of property values or other features.

5. Variance Inflation Factor (VIF):

VIF was calculated for each predictor to assess multicollinearity (high correlation between predictors).

Feature	VIF
Total_area	1.069984
latitude	12.670961
longitude	12.688553

Table 8: Variance Inflation Factor

Discussion: The Variance Inflation Factor (VIF) analysis was conducted to assess multicollinearity among predictor variables in our regression models. VIF quantifies how much the variance of the estimated regression coefficient is inflated due to multicollinearity. A VIF of 1 indicates no correlation among the predictor variable and the other predictor variables. Generally, VIF values exceeding 4 warrant further investigation, while VIFs exceeding 10 are signs of serious multicollinearity requiring correction.

In our analysis:

- total_area: The VIF value for Total area is 1.069984, indicating that there is no significant multicollinearity issue with this feature.
- lat (latitude): The VIF value for latitude is 12.670961, suggesting a high degree of multicollinearity. This implies that the information provided by latitude is highly correlated with one or more of the other predictor variables.
- lon (longitude): The VIF value for longitude is 12.688553, also indicating a high degree of multicollinearity. This is expected as latitude and longitude are inherently correlated geographical coordinates.

6. Principal Component Analysis (PCA)

- **Dimensionality Reduction:** PCA was applied to reduce the dimensionality of the dataset by transforming the original features into a smaller set of uncorrelated variables (principal components).

Discussion: PCA helped mitigate the issue of multicollinearity identified by the VIF analysis. The reduced feature set was then used for subsequent modelling steps. Given that our VIF analysis identified multicollinearity only between latitude and longitude, utilizing PC1 as a replacement for these two variables is a reasonable and effective strategy. It simplifies our model, addresses multicollinearity, and retains the most relevant geographical information for predicting property prices. We will proceed with this approach and evaluate the model's performance to ensure its effectiveness.

7. Polynomial Regression

- **Non-Linear Modelling:** Polynomial regression, a generalization of linear regression, was employed to capture potential non-linear relationships between features and property prices.
- **Model Evaluation:** The performance of the polynomial regression model was evaluated using the same metrics as before.

Discussion: The results of the polynomial regression model are somewhat mixed. While it outperforms the lasso regression model and shows potential for capturing non-linear relationships in the data, its performance is not significantly better than the linear or ridge regression models. This could be attributed to several factors:

- **Overfitting:** Polynomial regression models are prone to overfitting, especially with higher-degree polynomials. It is possible that our model is overfitting the training data, leading to a lower R-squared and higher MSE on unseen data.

- **Model Complexity:** Increasing the degree of the polynomial increases the model's complexity, which can make it harder to interpret and may not necessarily lead to better generalization.
- **Data Limitations:** The dataset itself might have limitations that hinder the performance of all models, including polynomial regression.

8. CONCLUSION

This project successfully developed a comprehensive real estate platform for the Nepalese market, combining a user-friendly marketplace built on the MERN stack with a robust Valuation Model (VM).

The MERN stack-based marketplace offers essential features such as property listings, search and filtering, in-app messaging, and bookmarking, enhancing the overall user experience for buyers, sellers, and agents.

The VM, a core component of the platform, was developed through data collection, pre-processing, analysis, and modelling. The initial linear regression model, while providing a baseline, highlighted the non-linearity and heteroscedasticity present in the data. Subsequent exploration of Ridge, Lasso, and polynomial regression, coupled with techniques like gradient descent, PCA, and model diagnostics, led to the selection of the Ridge Regression as the most suitable model for accurate property valuation.

Key findings from the project include:

- **Real Estate Marketplace:** The developed marketplace provides a user-friendly platform with features like property listings, search and filtering, in-app messaging, and bookmarking, which enhances the trust of stakeholders and eventually benefit the online property buying selling trend.
- **Data Challenges:** The absence of a centralized real estate database in Nepal presented a significant hurdle. However, a continuous and multi-pronged data collection approach and thorough cleaning and pre-processing can mitigate this.
- **Limited Sensitivity to Latitude and Longitude:** In the context of the Nepalese real estate market, latitude and longitude showed limited sensitivity in predicting property prices. This suggests that other location-based factors, such as proximity to amenities or specific neighbourhoods, might play a more significant role in determining property values.

- **Influence of Popular areas:** The dataset included property data from popular destinations, which tended to have higher property values compared to nearby areas. This introduced variance and potentially skewed the model's predictions for properties located in these specific regions.
- **Non-Linear Relationships and Heteroscedasticity:** The Rainbow Test and Breusch-Pagan test results confirmed the presence of non-linearity and heteroscedasticity in the relationship between property features and prices. This indicates that traditional linear regression models might not be the most suitable for capturing the complexities of the Nepalese real estate market.

These key findings highlight the unique challenges and opportunities in the Nepalese real estate market. The insights gained from this project will inform future research and development efforts, potentially leading to even more accurate and tailored property valuation models for this specific context.

9. FUTURE WORK

This project envisions a dynamic future roadmap for both the real estate marketplace and the Valuation Model (VM), aimed at continuous improvement, enhanced user experience

- **Broker Verification:** To enhance trust and credibility, we will implement a robust broker verification process during registration, ensuring that only legitimate and qualified professionals can list properties on the platform.
- **Enhanced Search and Filtering with K-Nearest Neighbours (KNN):** We will leverage KNN clustering, a machine learning technique, to improve the search and filtering capabilities of the marketplace.
- **Financial Planning and Advisory Section:** A dedicated section will be added to the marketplace to offer financial planning and advisory services related to real estate. This will include tools such as mortgage calculators, affordability assessments, and investment analysis, empowering users to make informed financial decisions.
- **Personalized Property Recommendations:** By leveraging user preferences, search history, and the insights gained from K-means clustering, we will develop a recommendation engine that suggests properties tailored to individual users' needs and preferences. This will enhance the user experience and streamline the property search process.
- **Continuous Model Improvement:** We will establish a data feedback loop by continuously collecting and integrating new property data from the marketplace into the Valuation Model's training dataset. This will ensure that the model remains up-to-date with the latest market trends and maintains its accuracy over time.
- **Local Model Customization:** We will leverage the neighborhoods clusters identified through K-means to develop localized AVM models. These models will be tailored to the specific characteristics of each neighborhoods, potentially leading to more accurate and precise valuations within those areas.
- **Advanced Machine Learning Techniques:** We will explore the integration of more advanced machine learning algorithms, such as gradient boosting, random forests, or deep learning, to further enhance the AVM's predictive power and accuracy.

- Explainable AI (XAI): We will incorporate XAI techniques to provide users with transparent and understandable explanations of the factors influencing property valuations. This will increase user trust in the AVM and promote informed decision-making.

By implementing these future enhancements, we envision the real estate marketplace and AVM evolving into a comprehensive and indispensable tool for the Nepalese real estate market. The platform will not only facilitate property transactions but also empower users with data-driven insights, personalized recommendations, and financial guidance, contributing to a more transparent, efficient, and accessible real estate ecosystem in Nepal.

10. PROJECT TASK AND TIME SCHEDULE

Increment	1		2		3					4	
Date	9 th May – 16 th May	16 th May – 23 rd May	23 rd May – 30 th May	30 th May – 6 th June	6 th June – 13 th June	13 th June – 20 th June	20 th June – 27 th June	27 th June – 4 th July	4 th July – 11 th July	11 th July – 19 th July	19 th July – 26 th July
Requirement Analysis											
Planning											
Documentation											
Coding											
Testing											

Index:

Completed Task	
Pending Task	

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APPENDIX

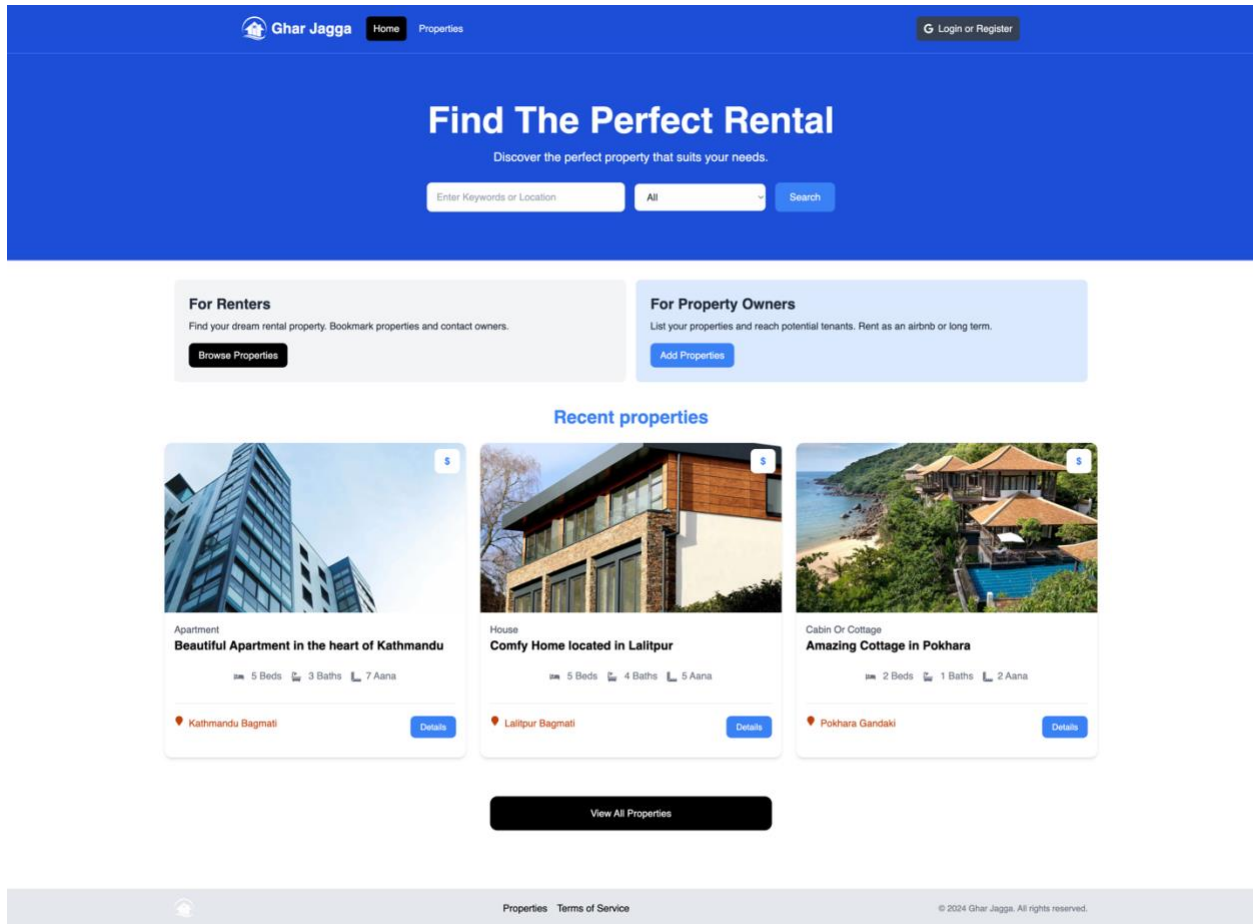


Figure 11: Home Page

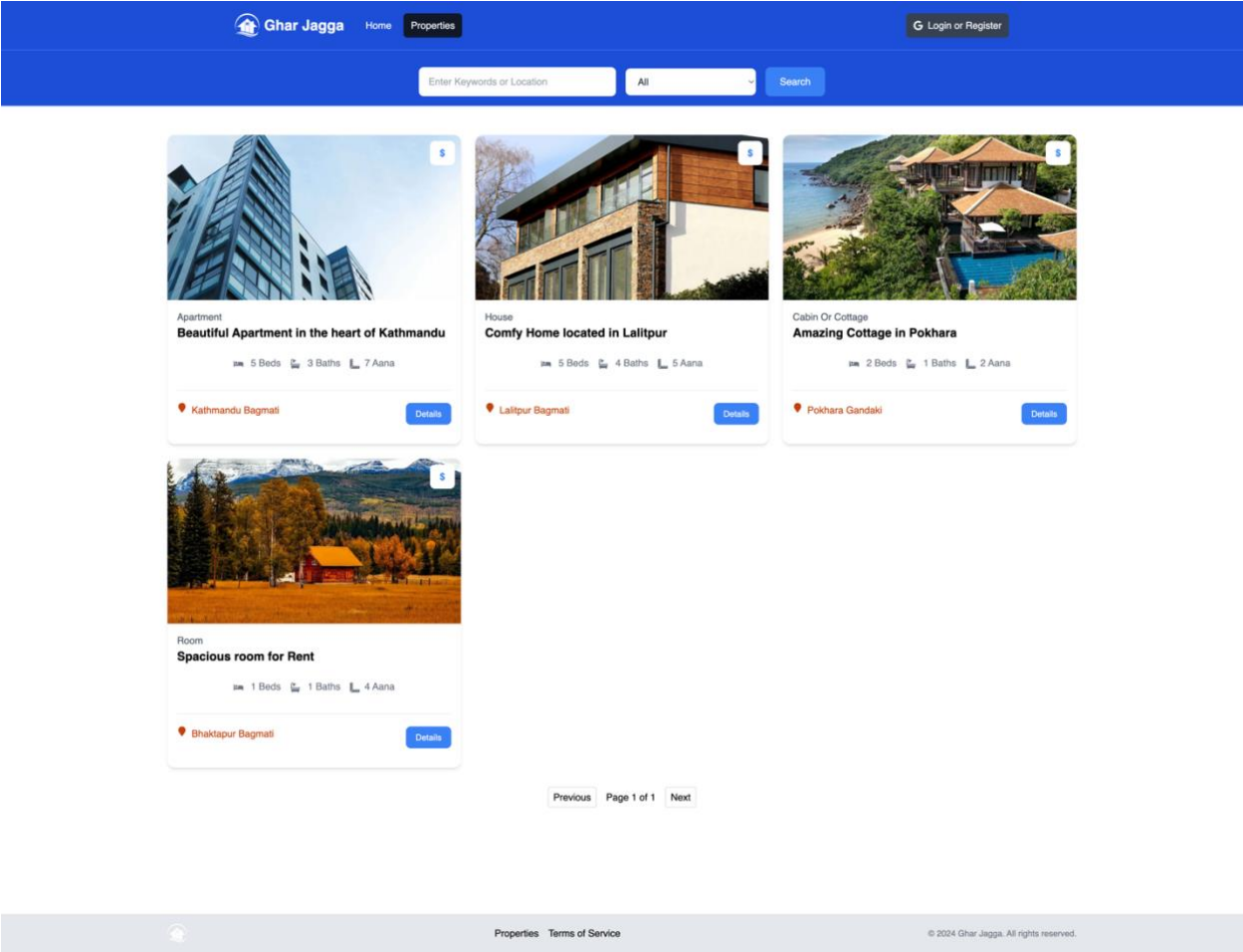


Figure 12: All Properties

Ghar Jagga

Home

Properties

Add Property





Add Property

Property Type

Apartment

Listing Name

eg. Beautiful Apartment in Kathmandu

Description

Add an optional description of your property

Location

Street

City

State

Zipcode

Bedrooms

Baths

Aana

Amenities

☐ Wifi

☐ Free Parking

☐ 24/7 Security

☐ Dishwasher

☐ Balcony/Patio

☐ Full kitchen

☐ Swimming Pool

☐ Wheelchair Accessible

☐ Gym/Fitness Center

☐ Smart TV

☐ Washer & Dryer

☐ Hot Tub

☐ Elevator Access

☐ Air Conditioning

☐ Coffee Maker

Rates (Leave blank if not applicable)

Weekly

Monthly

Nightly

Seller Name

Name

Seller Email

Email address

Seller Phone

Phone

Images (Select up to 4 images)

Choose files

No file chosen

Add Property



Properties

Terms of Service

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Figure 13: Add Properties Form

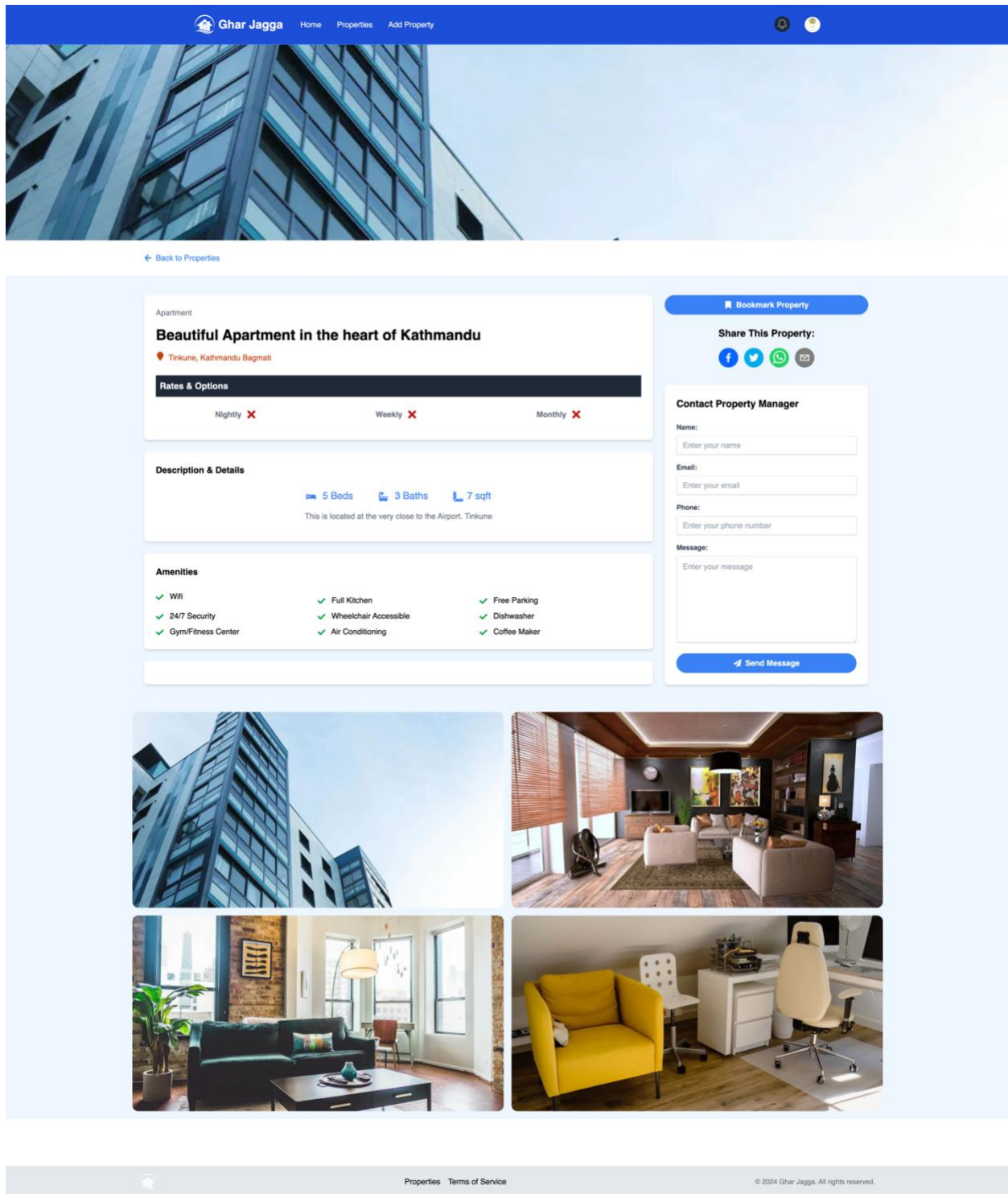


Figure 14: Single Property Detail

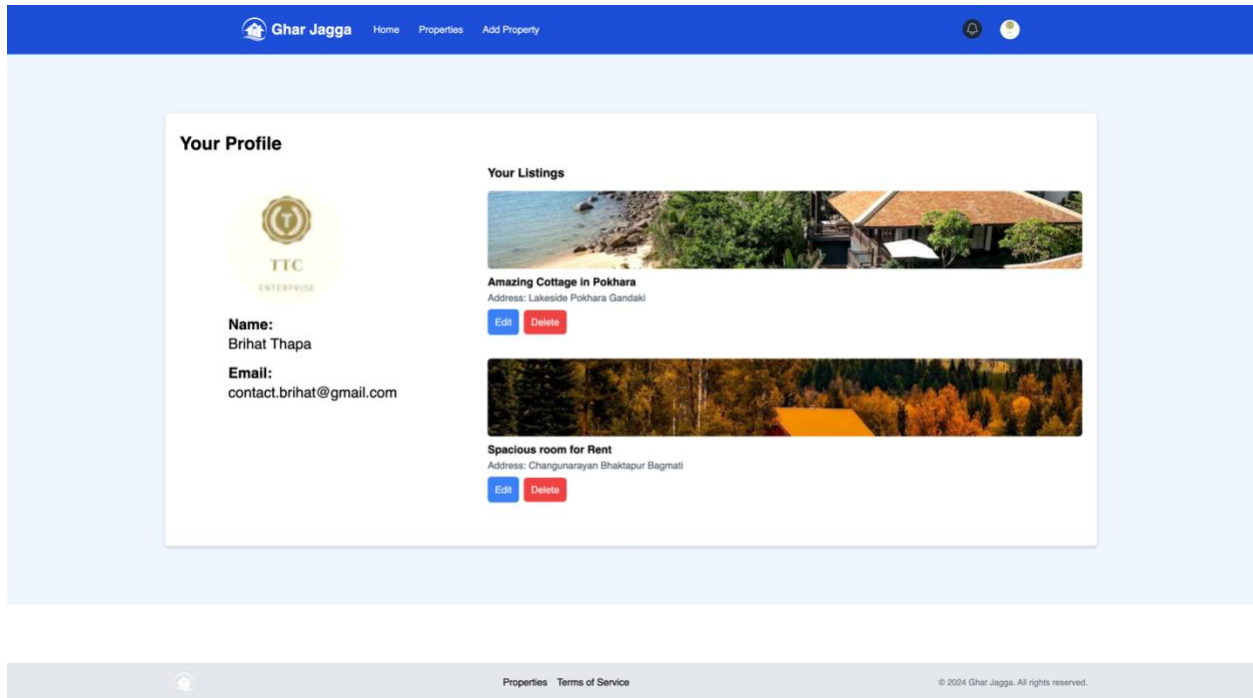


Figure 15: Saved Properties

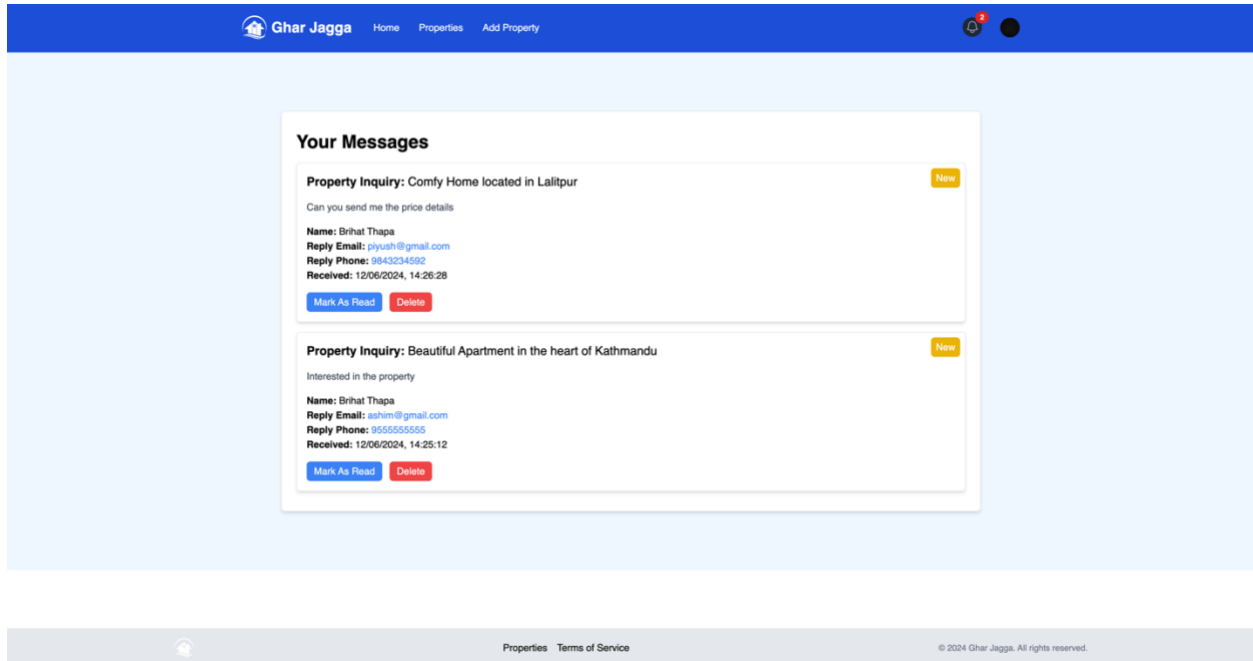


Figure 16: In App Communication

 Ghar Jagga

Home

Properties

Property Valuation

Add Property





Enter Property Details for Property Valuation

Property Type

Apartment

Location

City

State

Zipcode

Bedrooms

Baths

Aana

 **Disclaimer:** The Results of the Valuation Model are still *experimental*

Get Property Value

Figure 17: Value Calculation Form

 Ghar Jagga

Home

Properties

Property Valuation

Add Property





Enter Property Details for Property Valuation

Property Type

Apartment

Location

Lalitpur

Bagmati

44700

Bedrooms

3

 **Disclaimer:** The Results of the Valuation Model are still *experimental*

Get Property Value

The Property Valuation

The predicted price is: Rs 3,51,97,898

Figure 18: Property Valuation Result